

## SAFETY DATA SHEET

Creation Date 24-November-2010

Revision Date 25-December-2021

Revision Number 4

### 1. Identification

**Product Name** Dilauroyl peroxide

**Cat No. :** AC349740000; AC349740050; AC349741000; AC349745000

**CAS-No** 105-74-8

**Synonyms** Dodecanoyl peroxide; LPO; Laurox<sup>®</sup>4 S; Lauroyl peroxide

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

**Organic peroxides** Type D

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**  
Heating may cause a fire

**Precautionary Statements****Prevention**

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Keep/Store away from clothing/combustible materials

Keep only in original container

Wear protective gloves/protective clothing/eye protection/face protection

**Storage**

Store at temperatures not exceeding 30 °C/ 86 °F. Keep cool

Protect from sunlight

Store away from other materials

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component        | CAS-No   | Weight % |
|------------------|----------|----------|
| Lauroyl peroxide | 105-74-8 | >95      |

### 4. First-aid measures

|                                        |                                                                                                                              |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.              |
| <b>Skin Contact</b>                    | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention. |
| <b>Inhalation</b>                      | Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. Get medical attention.   |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Clean mouth with water. Get medical attention.                                                       |
| <b>Most important symptoms/effects</b> | No information available.                                                                                                    |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                        |

### 5. Fire-fighting measures

|                                       |                                                                              |
|---------------------------------------|------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>   | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. |
| <b>Unsuitable Extinguishing Media</b> | No information available                                                     |
| <b>Flash Point</b>                    | No information available                                                     |
| <b>Method -</b>                       | No information available                                                     |
| <b>Autoignition Temperature</b>       | >380 °C / >716 °F                                                            |
| <b>Explosion Limits</b>               |                                                                              |
| <b>Upper</b>                          | No data available                                                            |
| <b>Lower</b>                          | No data available                                                            |

**Oxidizing Properties**

Oxidizer

**Sensitivity to Mechanical Impact** No information available**Sensitivity to Static Discharge** No information available**Specific Hazards Arising from the Chemical**

Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood paper, oil, clothing, etc.).

**Hazardous Combustion Products**Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA****Health**  
0**Flammability**  
0**Instability**  
1**Physical hazards**  
N/A**6. Accidental release measures****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions**

See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Remove all sources of ignition. Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.

**7. Handling and storage****Handling**

Avoid contact with skin and eyes. Do not breathe dust. Do not subject to grinding/shock/friction. Keep away from clothing and other combustible materials.

**Storage.**

Do not store near combustible materials. Keep refrigerated. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep at temperature not exceeding 30°C. Incompatible Materials. Acids. Bases. Metals. Reducing Agent. Finely powdered metals. Strong reducing agents. Combustible material.

**8. Exposure controls / personal protection****Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies.

OSHA - Occupational Safety and Health Administration

**Engineering Measures**

Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**  
**Hand Protection**Goggles  
Protective gloves**Glove material**  
Natural rubber  
Nitrile rubber  
Neoprene  
PVC**Breakthrough time**  
See manufacturers  
recommendations**Glove thickness**  
-**Glove comments**  
Splash protection only

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

No information available.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                                    |                                                |
|----------------------------------------------------|------------------------------------------------|
| Physical State                                     | Powder Solid                                   |
| Appearance                                         | White                                          |
| Odor                                               | Slight                                         |
| Odor Threshold                                     | No information available                       |
| pH                                                 | No information available                       |
| Melting Point/Range                                | 53 - 57 °C / 127.4 - 134.6 °F                  |
| Boiling Point/Range                                | No information available                       |
| Flash Point                                        | No information available                       |
| Evaporation Rate                                   | Not applicable                                 |
| Flammability (solid,gas)                           | No information available                       |
| Flammability or explosive limits                   |                                                |
| Upper                                              | No data available                              |
| Lower                                              | No data available                              |
| Vapor Pressure                                     | No information available                       |
| Vapor Density                                      | Not applicable                                 |
| Specific Gravity                                   | No information available                       |
| Solubility                                         | No information available                       |
| Partition coefficient; n-octanol/water             | No data available                              |
| Autoignition Temperature                           | >380 °C / >716 °F                              |
| Decomposition Temperature                          | No information available                       |
| Viscosity                                          | Not applicable                                 |
| Molecular Formula                                  | C <sub>24</sub> H <sub>46</sub> O <sub>4</sub> |
| Molecular Weight                                   | 398.63                                         |
| Self-Accelerating Decomposition Temperature (SADT) | 50°C                                           |

## 10. Stability and reactivity

|                        |                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard        | Yes                                                                                                                         |
| Stability              | Oxidizer: Contact with combustible/organic material may cause fire.                                                         |
| Conditions to Avoid    | Keep away from open flames, hot surfaces and sources of ignition. Incompatible products. Combustible material. Excess heat. |
| Incompatible Materials | Acids, Bases, Metals, Reducing Agent, Finely powdered metals, Strong reducing agents, Combustible material                  |

**Hazardous Decomposition Products** Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>)

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Component Information

| Component        | LD50 Oral         | LD50 Dermal       | LC50 Inhalation |
|------------------|-------------------|-------------------|-----------------|
| Lauroyl peroxide | >2000 mg/kg (Rat) | >2000 mg/kg (Rat) | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component        | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------|----------|------------|------------|------------|------------|------------|
| Lauroyl peroxide | 105-74-8 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** Not mutagenic in AMES Test

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. .

| Component        | Freshwater Algae | Freshwater Fish                                         | Microtox   | Water Flea |
|------------------|------------------|---------------------------------------------------------|------------|------------|
| Lauroyl peroxide | Not listed       | LC50: > 0.3 mg/L, 96h semi-static (Poecilia reticulata) | Not listed | Not listed |

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

| Component        | log Pow |
|------------------|---------|
| Lauroyl peroxide | >6.5    |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN3106  
 Proper Shipping Name ORGANIC PEROXIDE TYPE D, SOLID  
 Technical Name Lauroyl peroxide  
 Hazard Class 5.2

#### TDG

UN-No UN3106  
 Proper Shipping Name ORGANIC PEROXIDE TYPE D, SOLID  
 Hazard Class 5.2  
 Packing Group II

#### IATA

UN-No UN3106  
 Proper Shipping Name ORGANIC PEROXIDE TYPE D, SOLID  
 Hazard Class 5.2

#### IMDG/IMO

UN-No UN3106  
 Proper Shipping Name ORGANIC PEROXIDE TYPE D, SOLID  
 Hazard Class 5.2

### 15. Regulatory information

#### International Inventories

| Component        | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Lauroyl peroxide | 105-74-8 | X   | -    | X    | ACTIVE                                        | 203-326-3 | -      | -   |

| Component        | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------|----------|-------|----------|------|------|------|------|-------|-------|
| Lauroyl peroxide | 105-74-8 | X     | KE-03388 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

#### Other International Regulations

## Authorisation/Restrictions according to EU REACH

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component        | CAS-No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Lauroyl peroxide | 105-74-8 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component        | CAS-No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Lauroyl peroxide | 105-74-8 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

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**Print Date** 25-December-2021

**Revision Summary** This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**